Truth Tables

We will now *define* the meaning of the propositional connectives. In doing this, the *semantics* of propositional logic will be formally unambiguously defined.

For each connective we will explicitly list its truth value depending on all possible combinations of truth values for its constituent propositions.

p	$\neg p$
F	**T**
T	**F**

p	q	$p \vee q$
F	**F**	**F**
F	**T**	**T**
T	**F**	**T**
T	**T**	**T**

p	q	$p \wedge q$
F	**F**	**F**
F	**T**	**F**
T	**F**	**F**
T	**T**	**T**

p	q	$p \Rightarrow q$
F	**F**	**T**
F	**T**	**T**
T	**F**	**F**
T	**T**	**T**

p	q	$p \Leftrightarrow q$
F	**F**	**T**
F	**T**	**F**
T	**F**	**F**
T	**T**	**T**

Truth Tables

Example: Consider the following statement:

Everyone who sits quietly for the next hour will get an ice cream.

This can be formalised as follows. Let Quiet stand for *"You sit quietly"*, and Ice stands for *"You get an ice cream"*. Then the above statement translates to $\mathrm{Quiet} \Rightarrow \mathrm{Ice}$ which has the following truth table:

Quiet	Ice	Quiet $\Rightarrow$ Ice
F	**F**	**T**
F	**T**	**T**
T	**F**	**F**
T	**T**	**T**

The *only* scenario in which the above statement can be considered false (third row of the truth table) is if you do not get an ice cream despite being quiet.

In particular, in the case where a noisy child gets an ice cream (second row of truth table) the statement is nonetheless true.

Truth Tables

Example: Catherine wishes to go to a party with either Jim or Jules. She is currently dating both, thus she doesn't want to go if they will both be there.

To analyse the scenario, let

- Cat stand for *"Catherine goes to the party"*
- Jim stand for *"Jim goes to the party"*
- Jules stand for *"Jules goes to the party"*

Catherine's predicament then can be formalised as $\mathrm{Cat} \Rightarrow \neg(\mathrm{Jim} \wedge \mathrm{Jules})$

That is, Catherine goes to the party *only if* Jim and Jules don't both go to the party.

We can build a truth table for this proposition to determine the scenarios in which it is true (i.e. when Catherine has no worries).

Truth Tables

We can determine the truth value of the proposition $\text{Cat} \Rightarrow \neg(\text{Jim} \wedge \text{Jules})$ by working out its truth table as follows.

Cat	Jim	Jules	$\text{Jim} \wedge \text{Jules}$	$\neg\,(\text{Jim} \wedge \text{Jules})$	$\text{Cat} \Rightarrow \neg\,(\text{Jim} \wedge \text{Jules})$
F	**F**	**F**	**F**	**T**	**T**
F	**F**	**T**	**F**	**T**	**T**
F	**T**	**F**	**F**	**T**	**T**
F	**T**	**T**	**T**	**F**	**T**
T	**F**	**F**	**F**	**T**	**T**
T	**F**	**T**	**F**	**T**	**T**
T	**T**	**F**	**F**	**T**	**T**
T	**T**	**T**	**T**	**F**	**F**

This tells us that the proposition is *false* if, and only if, all three participants in this love triangle go to the party.

Truth Tables

A more succinct way of filling the truth table is as follows:

Cat	Jim	Jules	Cat	$\Rightarrow$	$\neg$	(Jim	$\wedge$	Jules)
F	**F**	**F**	**F**	**T**	**T**	**F**	**F**	**F**
F	**F**	**T**	**F**	**T**	**T**	**F**	**F**	**T**
F	**T**	**F**	**F**	**T**	**T**	**T**	**F**	**F**
F	**T**	**T**	**F**	**T**	**F**	**T**	**T**	**T**
T	**F**	**F**	**T**	**T**	**T**	**F**	**F**	**F**
T	**F**	**T**	**T**	**T**	**T**	**F**	**F**	**T**
T	**T**	**F**	**T**	**T**	**T**	**T**	**F**	**F**
T	**T**	**T**	**T**	**F**	**F**	**T**	**T**	**T**
			(1)	**(4)**	(3)	(1)	(2)	(1)

The numbering along the bottom indicates the order in which the entries are added:

(1) Values for variables first

(2) The most tightly-binding operation ($\wedge$) comes next

(3) The next operation ($\neg$) can then be filled in

(4) The outermost operation ($\Rightarrow$) can then be inserted

Truth Tables

Exercise: Construct the truth table for $\neg(P \Leftrightarrow \neg Q)$

P	Q	¬	(	P	⇔	¬	Q	)
F	**F**							
F	**T**							
T	**F**							
T	**T**							

Exercise: How many rows will there be in a truth table involving four propositional variables P, Q, R and S?

What if there are five propositional variables?

What if there are n propositional variables?

Tautologies and Contradictions

A formula p is:

1. a ***tautology*** if, and only if, it is true under all interpretations.
   In this case we say p is ***valid***.
2. a ***contradiction*** if, and only if, it is false under all interpretations.
   In this case we say p is ***unsatisfiable***.
3. ***satisfiable*** if, and only if, it is true under some interpretations;
   i.e., it is *not* a contradiction.

Two formulas p and q are ***logically equivalent*** if $\mathrm{p} \Leftrightarrow \mathrm{q}$ is a tautology; i.e., if their truth tables produce the same truth values.

Example: $\mathrm{p} \vee \neg \mathrm{p}$ is a tautology, $\mathrm{p} \wedge \neg \mathrm{p}$ is a contradiction, and $\neg \neg \mathrm{p}$ is equivalent to p, which can be formally verified by constructing their truth tables:

p	$\neg \mathrm{p}$	$\neg \neg \mathrm{p}$	$\mathrm{p} \vee \neg \mathrm{p}$	$\mathrm{p} \wedge \neg \mathrm{p}$
F	**T**	**F**	**T**	**F**
T	**F**	**T**	**T**	**F**

Tautologies and Contradictions

Exercise: Construct truth tables for each of the following formulas to determine which are tautologies and which are contradictions.

1.

p	q	$p \vee (\neg p \wedge q)$
F	**F**	
F	**T**	
T	**F**	
T	**T**	

2.

p	q	$(p \wedge q) \wedge \neg(p \vee q)$
F	**F**	
F	**T**	
T	**F**	
T	**T**	

3.

p	q	$p \Rightarrow (q \Rightarrow p)$
F	**F**	
F	**T**	
T	**F**	
T	**T**	

4.

p	q	$(p \Rightarrow \neg p) \Leftrightarrow \neg p$
F	**F**	
F	**T**	
T	**F**	
T	**T**	

Equivalences and Valid Arguments

Tautologies are important in ascertaining the validity of arguments, such as:

1. Either this man is dead or my watch has stopped.
2. My watch is still ticking.
   Therefore
3. This man is dead.

This argument is valid if the two premises imply the conclusion; that is, if:

$$(\,\text{Dead} \vee \text{Stopped}\,) \wedge \neg\text{Stopped} \Rightarrow \text{Dead}$$

is a tautology. We can confirm this by constructing its truth table:

Dead	Stopped	(Dead ∨ Stopped)			∧	¬ Stopped		⇒	Dead
F	**F**	**F**	**F**	**F**	**F**	**T**	**F**	**T**	**F**
F	**T**	**F**	**T**	**T**	**F**	**F**	**T**	**T**	**F**
T	**F**	**T**	**T**	**F**	**T**	**T**	**F**	**T**	**T**
T	**T**	**T**	**T**	**T**	**F**	**F**	**T**	**T**	**T**
		(1)	(2)	(1)	(3)	(2)	(1)	(4)	(1)

Equivalences and Valid Arguments

In contrast, consider the following argument:

1. Barking dogs don't bite.
2. My dog doesn't bark. *Therefore*
3. My dog bites.

Its formalisation yields the following truth table:

Barks	Bites	(Barks	⇒	¬ Bites)	∧	¬ Barks	⇒	Bites
F	**F**	**F**	**T**	**T**	**F**	**T**	**T**	**F**	**F**	**F**
F	**T**	**F**	**T**	**F**	**T**	**T**	**T**	**F**	**T**	**T**
T	**F**	**T**	**T**	**T**	**F**	**F**	**F**	**T**	**T**	**F**
T	**T**	**T**	**F**	**F**	**T**	**F**	**F**	**T**	**T**	**T**
		(1)	(3)	(2)	(1)	(4)	(2)	(1)	(5)	(1)

The first row of this truth table shows that the proposition – and hence the argument it represents – is not valid. That is, it presents a scenario in which the proposition may be false: we may have a dog that neither barks nor bites.